COMPANION TO THE MERIT SPECIFICATION

MERIT screen mock-ups

Eleven screens from the Metals Exposure Registry and Investigation Tracking specification, showing how its data rules appear in daily work: intake, matching, case management, field investigations, configuration and federal reporting.

Draft for review
October 24, 2026
12 pages · landscape Letter

How to read a page

1 Numbered markers on each screen match the notes printed beneath it. Each note says which rule the element carries out; the page header lists the spec sections the screen draws on.

Every screen is drawn 1180 pixels wide, a tablet held in landscape, because the spec asks for layouts that work at tablet width for field staff. Data-heavy screens show one state of a scrolling page; content below the fold is cut off as it would be on the device.

Design basis

- **USWDS 3 without its JavaScript.** Styles, markup patterns and design tokens come from USWDS; dialogs, menus, combo boxes and date pickers are Angular CDK components wearing USWDS classes.
- **One shared data grid** for work queues, held records, results and extract reports: server-side paging, sorting and filtering, saved views, bulk selection and keyboard navigation.
- **WCAG 2.1 AA.** Status always pairs color with an icon or a word, field-by-field marks spell out their meaning, and every action is reachable from the keyboard.
- **Rules are data.** Thresholds, standards, schedules and extract layouts are effective-dated rows, so screens show where a level came from and when it took effect.

All data is synthetic. People, addresses, phone numbers (555-01xx), case numbers, labs, clinics, employers and results are invented for illustration. NPI 1234567893 and CLIA 99D1234567 are format examples only. Maryland counties appear because the spec uses Maryland for its examples; no agency's data or branding is shown.

Screens

1	Home dashboard and work queues	urgent holds, queues, batches	2
2	Person page: child	places, results trend	3
3	Set as primary residence	row split, county change	4
4	Person page: adult worker	employment first, ABLES status	5
5	Result entry	cascading fields, save summary	6
6	Held records	fix the cause, release the rows	7
7	Person matching review	field-by-field comparison	8
8	Intake profile builder	mapping, translations, preview	9
9	Case and investigations	workflow, sites, samples	10
10	Threshold administration	tiers, authorities, effective dates	11
11	Federal extracts	CBLS and ABLES validation	12

People in the examples

Maya Torres, 4	Confirmed lead case; splits time between parents' homes
Leo Torres, 2	Maya's brother; new capillary result entered on screen 5
Ana and Daniel Torres	Parents and guardians; mother's home is the primary residence
Marcus Bell, 38	Lives in the mother's home; works at a battery recycler
JON A BAKER	Incoming lab record waiting for a person match

Legend

Status: ! Urgent · ⚠ Needs attention · ⌛ In progress · ✓ Done

Match marks: ✓ Agrees · ≈ Close · × Different · — Missing

Home dashboard and work queues

SPEC SECTIONS
User interface › Work queues · Intake › Held records, Intake pipeline

https://merit.example.org/home — 1180 px wide · tablet landscape

M MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | JR Jamie Rivera, State lead program | 5

Home

Scope: State lead program and all 24 sub-jurisdictions

Saturday, 10/24/2026 | Arrange queues

1

! Urgent: held results at or above an action level 2 — Open held records

48 µg/dL Lead · capillary
Missing DOB · R. NGUYEN (as reported) · collected 10/23/2026 · Bayline Reference Lab, batch B-26-10240
No usable DOB, so the lowest action level for any population applies; 48 µg/dL is in the ≥ 45 tier: 48 hours
Waiting 3 h | Open

22 µg/dL Lead · capillary
Uncertain person match · TYLER BROOKS, DOB 01/17/2023 · collected 10/22/2026 · Fernwood Family Clinic
Child tier 20–44 µg/dL: confirmatory venous test due within 2 weeks
Waiting 5 h | Review match

Work queues 2

Counts open the shared data grid, filtered to the queue

23 Unmatched results — 19 with a suggested match · oldest 2 days	**5** Unknown LOINC codes or units — 2 senders · 4 rows wait on code PBVEN	3 **1** Paused batches — Bayline Reference Lab CSV: column headers changed	**12** Address review — New lab addresses that differ from the primary	**9** Unverified primary residences — On open cases · lab-reported address
4 Detection limit above case level — By reporter · Fernwood Family Clinic 3	**3** Pregnancy episodes past expected end — Oldest 21 days past	**7** Cases with overdue steps — By worker · Dana Wells 3, Sam Okafor 2	**4** Cross-link and cluster tasks — New cluster: 1400 block of Maple Ridge Ct	**31** Missing extract-required fields — Before CBLS 2026 Q3 and ABLES 2026

4

Today's intake batches

All batches

Sender	Recv.	Loaded	Held	Rej.
Bayline Reference Lab HL7 ORU · 06:12	412	398	14	0
Fernwood Family Clinic CSV · 07:05	38	37	1	0
Bayline Reference Lab CSV · 07:40	Paused: layout changed — 156 rows wait; nothing loaded yet			
eICR feed FHIR · 08:15	9	9	0	0

Federal submissions

Extracts

CBLS 2026 Q3 — Jul–Sep · validated with 3 warnings — Review
ABLES 2026 — Dry run · 4 rules failing — 10 records

1 Urgent holds come first
A held result at or above an action level appears here, not only in its queue. With no usable DOB, the lowest action level for any population applies.
Held records › Urgent first · Population derivation 1

2 The ten queues from the spec
Each tile shows a count and the breakdown the spec names (by reporter, by worker). A count opens the shared data grid with that queue's saved view.
User interface › Work queues

3 Paused, not failed
A sender's header signature changed, so the whole batch stopped before loading. It resumes once someone updates the mapping (screen 8).
Intake profiles › Layout drift

4 Batches never fail whole
Each batch reports received, loaded, held and rejected rows, and each count links to the rows themselves.
Intake pipeline

5 Scope is visible
Counts cover the signed-in user's tenant and everything below it. Row-level security applies the same scope to every query and export.
Architecture › Tenant tree, Row-level security

Person page: child

SPEC SECTIONS
User interface › Person page, Places behavior · Address rules 6 and 11

https://merit.example.org/people/P-0041-2261 — 1180 px wide · tablet landscape

M MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | DW Dana Wells, Baltimore City LHD

Maya Torres CHILD (1)

+ Add result | Add place | ···

DOB 03/14/2022 · 4 y 7 m · Female · MERIT ID P-0041-2261 · 1 alias · Lead · Confirmed · Open · Primary residence: **Baltimore City** · Case manager: Dana Wells

Overview | Relationships 4 | Places 4 | Results 4 | Cases and investigations 1 | Employment 0 | Exposure sources 0 | Pregnancy 0 | Timeline (2)

Relationships — Add relationship

Ana Torres	Mother · guardian	Primary contact
Daniel Torres	Father · guardian	Lives at father's home
Leo Torres	Brother · 2 y	Lead · Suspect · Open
Marcus Bell	Household member	Lead · Confirmed · Open

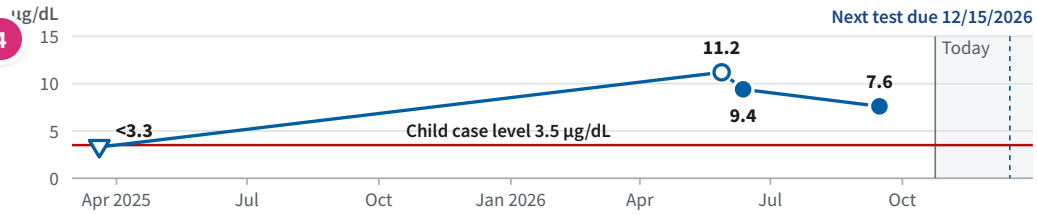
Places — Investigation lookback: 12 months

★ **Mother's home**	Residence · Primary · custody order 1412 Maple Ridge Ct, Baltimore, MD 21218 · Baltimore City	Weekdays since 08/01/2024
☆ **Father's home**	Residence · Set as primary… 507 Cedar Hollow Rd, Bel Air, MD 21014 · Harford County	Weekends since 08/01/2024
ABC Child Care	Child care 220 Willow Bend Rd, Baltimore, MD 21218 · Baltimore City	Weekdays 40 h/week
Grandmother's home	Relative or sitter home 31 Orchard Ln, Baltimore, MD 21229 · Baltimore City ⚠ 8 h/week is under the 10 h minimum: review task	2 afternoons 8 h/week

(3)

Results — Lead · blood — Export

(4)



Collected ↓	µg/dL	Sample	Method	Reporter	Effect
09/15/2026	**7.6**	Venous	ICP-MS	Bayline Reference Lab	Follow-up
06/12/2026	**9.4**	Venous	ICP-MS	Bayline Reference Lab	Confirmed the case
05/28/2026	**11.2**	Capillary	Point-of-care	Fernwood Family Clinic	Opened suspect case
03/20/2025	**<3.3**	Capillary	Point-of-care	Fernwood Family Clinic	Below detection

Lead case C-26-00872 · Confirmed · Open
Child residential workflow, step 5 of 8 · 3 investigations · 1 review task — Open case

(5)

Employment — No jobs recorded

Exposure sources — None recorded yet

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1 Header
Name, DOB, age, population badge, one chip per open case and analyte, and the county of the current primary residence. Aliases are one click away.
Person page › Header

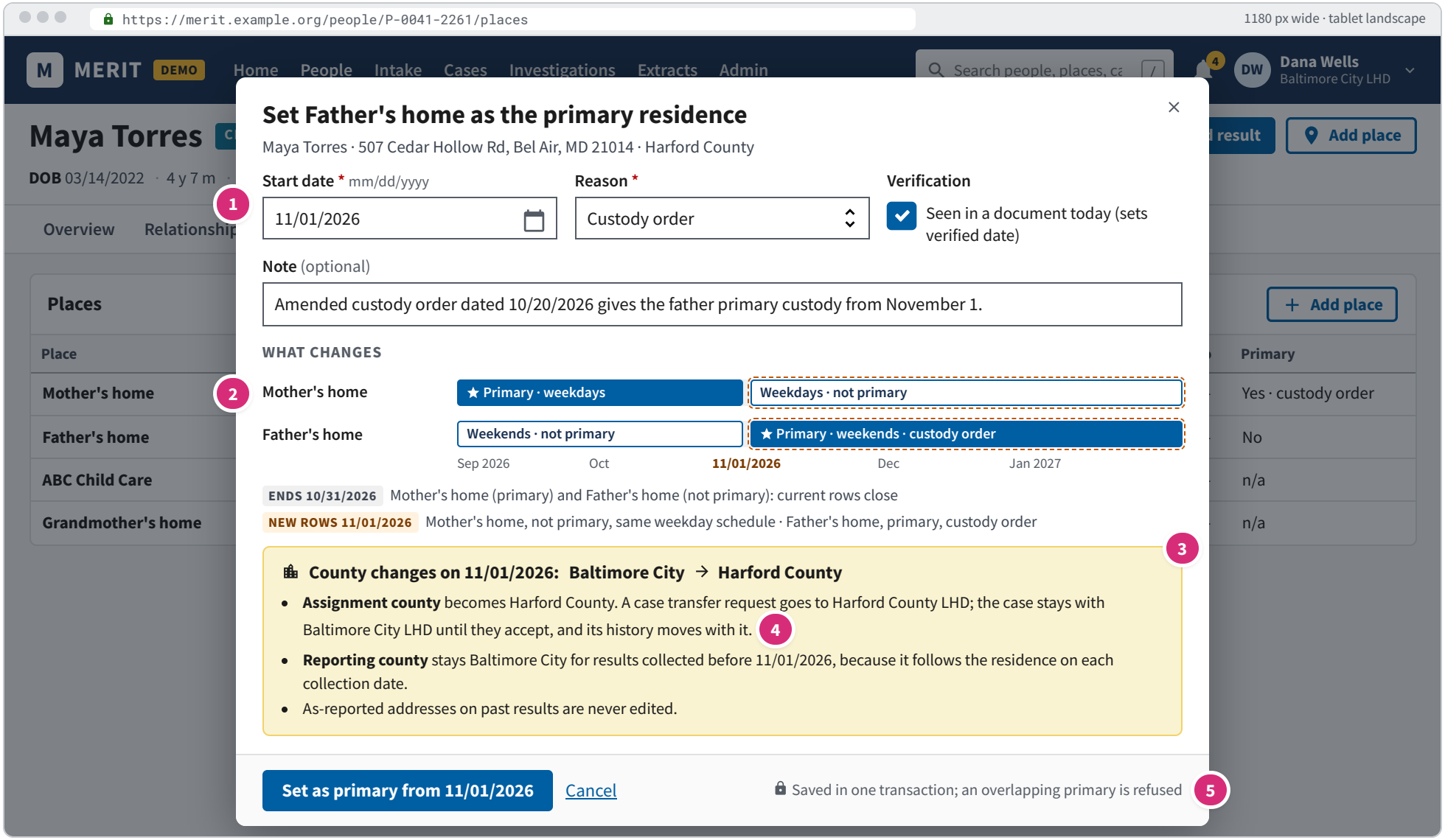
2 Order follows population
For a child, Relationships and Places come before Results; an adult page leads with Employment (screen 4). Empty sections stay in the list with a zero.
Person page › section order

3 Places carry the rules
The primary star shows its reason; "Set as primary" opens screen 3. The grandmother's home is under the 10-hour minimum, so it gets a review task instead of an investigation request.
Places behavior · Address rule 11

4 Trend chart
The case level comes from the threshold table, venous and capillary results use different markers, and a "<3.3" result sits at its detection limit. The table below carries every value.
Person page › Results · Lab result › operator

5 Empty states
A section with nothing in it says so ("No jobs recorded") instead of disappearing, so staff can tell missing data from a hidden section.
Person page › empty sections

Set as primary residence



1 Reason and start date
Changing the primary always asks why and from when. Typical reasons are a custody order, most overnights or a guardian's designation.
Places behavior · Address rule 3

2 Rows split, history stays exact
The old primary ends the day before the start date. Because Maya still lives at her mother's home, a new non-primary row continues there with the same schedule.
Address rule 12

3 Two counties, two jobs
The dialog shows the county change before saving. Assignment follows the current primary; reporting follows the residence on each collection date, so past counts do not move.
Address rules 6 and 7

4 Transfer, not silent move
A child moving to another county moves the case by a transfer request that the receiving tenant accepts. The person stays one statewide record.
Cross-tenant work · Data ownership

5 Enforced below the UI
The stored procedure that saves person locations locks the person's rows and refuses overlapping primaries, so imports and the API obey the same rule.
Address rule 1

Person page: adult worker

https://merit.example.org/people/P-0038-1187 — 1180 px wide · tablet landscape

MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | RC Rosa Chen, Occupational health program

Marcus Bell ADULT

+ Add result | Add job | ···

DOB 08/30/1988 · 38 y · Male · MERIT ID P-0038-1187 | Lead · Confirmed · Open | Primary residence: **Baltimore City** | Investigator: Rosa Chen

Overview | Employment 2 | Places 1 | Results 3 | Cases and investigations 2 | Relationships 3 | Exposure sources 0 | Timeline | 1

Take-home exposure review: lives at 1412 Maple Ridge Ct with Maya Torres (confirmed) and Leo Torres (suspect). Assigned to State lead program. Open task 5

Employment 2

2 jobs · 1 current · Show past job

Harbor Battery Recycling LLC Current since 03/2019 · 45 h/week | Verified 04/22/2026

Worksite	2200 Tidewater Industrial Pkwy, Baltimore, MD 21226 · Baltimore City
As stated	Job: furnace operator · Industry: battery recycling, lead smelting
NAICS 2022	331492 Secondary smelting, refining and alloying of nonferrous metal (e…
Occupation (COC)	Not coded	2 suggested codes	Code occupation 3
Metal tasks	Charges and taps the furnace; breaks battery cases
PPE	Half-face respirator (P100); change room on site; wears work boots home

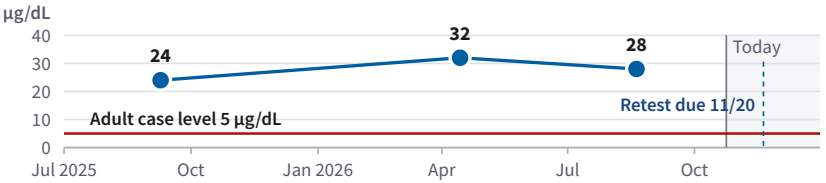
Cases and investigations

Adult occupational workflow

- **Lead case C-26-00511 · Confirmed · Open** — Exposure category: occupational · step 4 of 7, code the work — Retest due 11/20
- **Workplace investigation · Harbor Battery Recycling** — Referred to state plan (MOSH) 05/04/2026 · shared by 3 workers' cases — Outcome pending

Results

Lead · blood



Collected ↓	µg/dL	Sample	Reason	Reporter
08/20/2026	**28**	Venous	Occupational monitoring	Bayline Reference Lab
04/14/2026	**32**	Venous	Occupational monitoring	Bayline Reference Lab
09/09/2025	**24**	Venous	Occupational monitoring	Bayline Reference Lab

ABLES 2026 status 4 — computed by NIOSH's rule, not the case level

Highest 2026 result, 04/14/2026 — **32 µg/dL · at or above 10**

In the 2025 registry at or above 10 — **Yes · 24 µg/dL**

Status 2 · Existing case | WorkRel 1 · work-related

1 Adult order
Employment comes right after the header and Relationships drops to sixth. Pregnancy is hidden because the person is recorded as male with no episodes.
Person page › section order

2 Employment, not a place
The worksite lives on the job row, with job title and industry kept as stated for coding, metal tasks, PPE and hours. NAICS comes from the employer unless this job overrides it.
Employment · Address rule 8

3 Code the work
The occupation still needs a Census code. An optional autocoder suggests codes for staff to accept; until then ABLES validation flags the record (990 means unknown).
Adult: occupational workflow, step 4

4 ABLES status is its own rule
The extract compares the year's highest result with 10 µg/dL and checks the prior year's registry, so Marcus is code 2, existing case, whatever the state's case level.
Adult lead · Federal extracts

5 Cross-link task
An adult case sharing a home with children's cases raises a take-home exposure review. It spans two tenants, so it goes to their nearest common ancestor.
Cluster detection · Cross-tenant work

Result entry

https://merit.example.org/people/P-0041-2262/results/new

1180 px wide · tablet landscape

M MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | CL Chris Lam, State lead program

People › Leo Torres › Results

Add result

Leo Torres CHILD · DOB 07/02/2024 · 2 y 3 m · Male · P-0041-2262 · Baltimore City

1 TEST

Analyte * : Lead
Specimen * : Whole blood
Unit * : µg/dL
Sample type * : Venous | Capillary | Unknown

Speciation and creatinine are hidden: lead in whole blood uses neither. Units list the canonical unit first.

2 RESULT AND DATES

Result * : = 12.4
Collected * : 10/22/2026
Analyzed : 10/22/2026
Reported : 10/22/2026

3 METHOD, DEVICE AND KIT

Test method : Point-of-care analyzer
Device model : PX-200
Device serial : 44871-B
Test kit lot : K2609-118

4 SOURCE

Lab or testing site * : Fernwood Family Clinic
CLIA 99D1234567 · certificate active
Ordering provider : Elena Marsh, MD
Linked by NPI 1234567893
Accession number : FFC-26-0412233
Reason for test : Screening
Payer : Medicaid
Result status : Final

5

Saved 10/24/2026 9:42 AM

Canonical value
12.4 µg/dL reported as 12.4 µg/dL

Child at collection
Age 2 y 3 m on 10/22/2026

Thresholds crossed
- Reportable: any value
- Child case level, 3.5 µg/dL
- Capillary tier 10–19 µg/dL: confirmatory venous test within 1 month

Suspect lead case opened · C-26-00931
One capillary result at or above the case level. Task: venous test due **11/22/2026**, assigned to Dana Wells.

Joined an open investigation
1412 Maple Ridge Ct already has a residential investigation for Maya Torres; Leo's case was linked to it instead of opening a second one.

Go to case | Add another result

1 Fields cascade
Specimens are filtered to the analyte, units to the pair with the canonical unit first. Sample type appears only because lead in blood uses it; speciation and creatinine stay hidden.
Result entry form, items 1–6

2 Value with operator
"<" is a real choice, not text: a "<3.3" result is stored as below detection with 3.3 as the limit. Collected must fall between DOB and the reported date.
Lab result › operator, collected_date

3 Method, device and lot
Recorded on every result so a recall query can find each affected result and patient. The method also decides whether a venous result can confirm a case.
Test method, device and lot

4 Linked by identifier
The provider links by NPI and the testing site by CLIA, from the shared cache; the name as typed stays in the result's snapshot.
Provider entity · Lab result

5 Save shows the consequences
The canonical value and every threshold crossed, the suspect case it opened with the due date from its tier, and the existing investigation it reused.
Result entry form · Child lead · Investigation

Held records

https://merit.example.org/intake/held

1180 px wide · tablet landscape

MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | CL Chris Lam, State lead program

Intake › Held records

Held records

41 rows · 9 causes · oldest waiting 4 days

Export | Columns | Save view

Group: **cause › sender › batch** (1) | Cause: **All** | Owner: **Anyone** | 1 selected group

	Patient, as reported	DOB	Test, as reported	Value	Collected	Waiting ↓	Owner
!	**At or above an action level** 2 sorted first, whatever the cause (2)						
☐	**NGUYEN, R** Missing DOB	—	PBCAP lead, cap.	**48**	10/23/2026	3 h	Chris Lam
☐	**BROOKS, TYLER** Uncertain person match	01/17/2023	10368-9 lead, cap.	**22**	10/22/2026	5 h	Chris Lam
☐	**Unknown test code PBVEN** Bayline Reference Lab · batch B-26-10241 4						
☑	GARCIA, SOFIA	02/11/2021	PBVEN	6.3	10/21/2026	1 d	Chris Lam
☑	PATEL, ARJUN	09/04/2022	PBVEN	2.1	10/21/2026	1 d	Chris Lam
☑	COLE, JORDAN	12/19/1990	PBVEN	4.0	10/20/2026	1 d	Chris Lam
☑	WILLIAMS, KEISHA	05/27/2019	PBVEN	3.6	10/20/2026	1 d	Unassigned
›	**Unknown unit PPB** Fernwood Family Clinic · batch F-26-0388 2						
›	**Unusable address** 3 senders 8						
›	**Possible duplicate result** 2 senders 6						

6 rows shown · 41 held · 6 groups collapsed

(5) Assign... | Reject...

Map local test code

(4)

As reported: PBVEN "LEAD VENOUS BLD" · Bayline Reference Lab

Held rows: 4 in batch B-26-10241; values stay as sent

LOINC code *

77307-7 · Lead [Mass/volume] in Venous blood

77307-7	Lead [Mass/volume] in Venous blood
5671-3	Lead [Mass/volume] in Blood
10368-9	Lead [Mass/volume] in Capillary blood

Implies analyte lead, specimen whole blood, sample type venous.

☑ Save to the sender's profile: always map PBVEN to 77307-7 for Bayline Reference Lab (3)

Releasing reruns validation and matching

- 3 rows load and classify
- 1 row (COLE, JORDAN) goes to person matching

Map and release 4 rows | Cancel

(1) **Grouped by cause**
Holds group by reason, then sender and batch, so one fix covers every row with the same problem. Every hold names its reasons from a fixed list.
Held records › Specific reasons, Grouped by cause

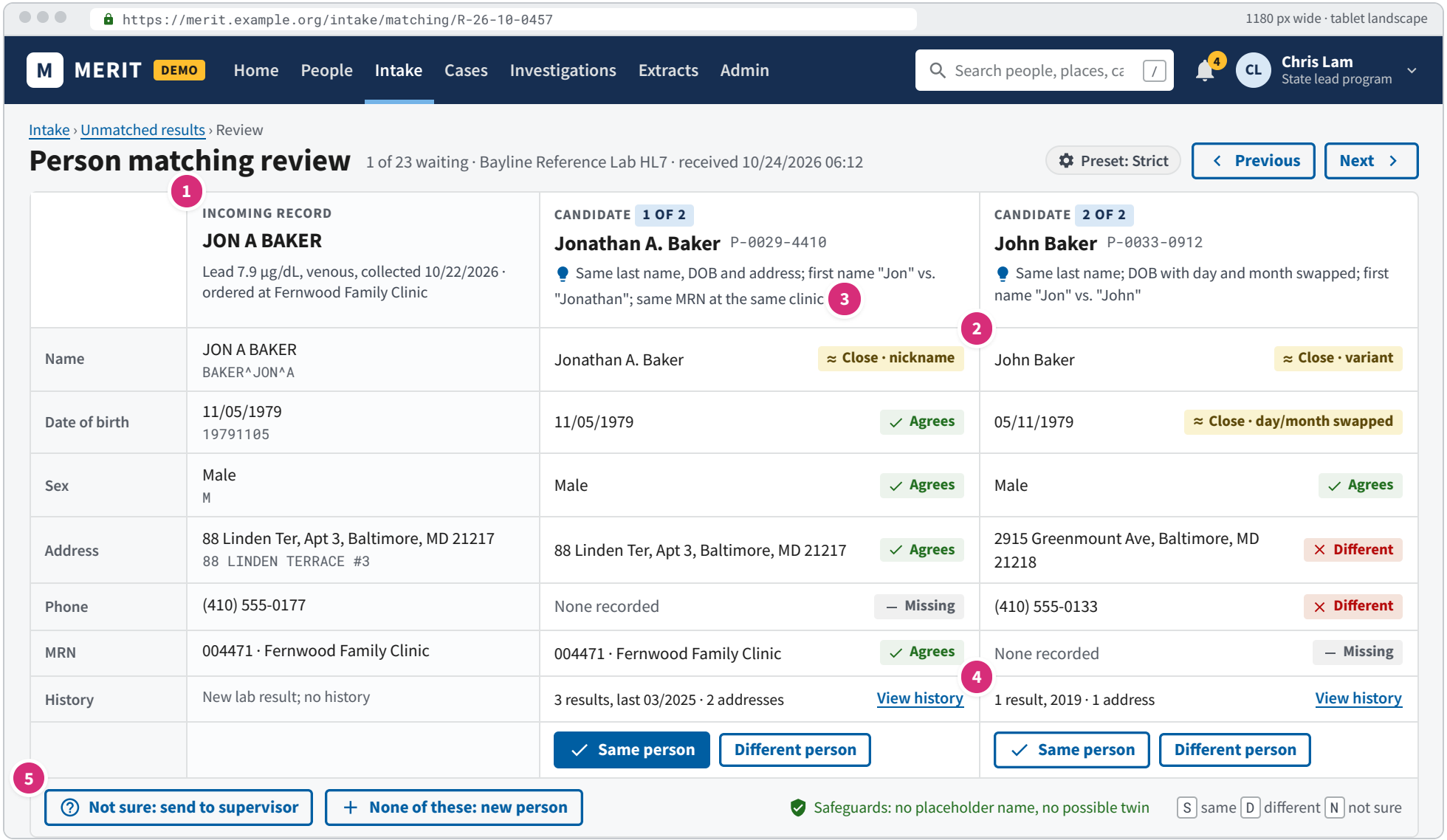
(2) **Urgent first**
Rows at or above an action level sort above every group. NGUYEN has no DOB, so the lowest action level for any population was applied.
Held records › Urgent first

(3) **Fix once, remember it**
Saving the mapping to the sender's intake profile means the next file from Bayline loads PBVEN without a hold.
Held records › Fix once · Intake profiles

(4) **Correction beside the original**
The code as sent stays visible next to the mapping. Corrections are logged; the result's as-reported snapshot never changes.
Held records › Edit in place

(5) **Owners, age, release, reject**
Holds can be assigned and show how long they have waited. Release reruns validation and matching; reject needs a reason and keeps the row.
Held records › Ownership, Release and reject

Person matching review



1 Side by side, field by field
The incoming record sits next to up to five ranked candidates. Values are shown standardized, with the raw value as sent underneath.
Person matching › Review screen

2 Marks say what they mean
Each field is marked agrees, close (a nickname, or day and month swapped), different or missing, with a word and an icon rather than color alone.
Person matching › Review screen

3 Why it was suggested
One plain line per candidate. Scores stay hidden; administrators pick a preset such as Strict and switch plain-language rules.
Person matching · Matching configuration

4 History one click away
A candidate's results and address history open in a side sheet without leaving the review.
Person matching › Review screen

5 Three plain outcomes
Same person links the result. Different person is remembered, so the pair is never suggested again. Not sure goes to a supervisor. Placeholder names and possible twins never link automatically.
Person matching › Built-in safeguards

Intake profile builder

https://merit.example.org/intake/profiles/fernwood-family-clinic/v1 — 1180 px wide · tablet landscape

MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | CL Chris Lam, State lead program

Paused batch: Bayline Reference Lab CSV, 10/24 07:40, has 2 renamed columns and 1 new one. 156 rows wait; none loaded. **Update that mapping** (1)

Intake › Intake profiles › New

Fernwood Family Clinic

Draft · version 1 — Sample: lead_results_oct.csv · 38 rows · comma-delimited · header row 1

Map columns (2)

Suggestions from header names and content · 14 columns, 9 shown

Column in file	Sample values	Maps to	Translation or note
PatientLast	TORRES · NGUYEN · HAYES	Patient last name	Matched header name
DOB	07/02/2024 · 01/30/2023	Date of birth	Format MM/DD/YYYY, detected
Sex	M · F	Sex	M → Male; F → Female
Zip	21218 · 21217 · 21229	ZIP code	Recognized from content
TestCode	PB-CAP	Local test code	PB-CAP → LOINC 10368-9 capillary
Result	12.4 · <3.3 · <5	Value with operator	"<" becomes operator and limit
Units	ug/dL	Unit	ug/dL → µg/dL
ProvNPI	1234567893	Ordering provider NPI	✓ Check digit valid
Guardian	ANA TORRES	Guardian name	Creates an unverified contact person

(3) **Constants for values the file leaves out:** lab = Fernwood Family Clinic (CLIA 99D1234567) · test method = point-of-care analyzer · specimen = whole blood

Preview (4)

First 5 rows, parsed and validated

Patient	Value	Result of checks
TORRES, LEO	12.4	✓ Ready
NGUYEN, ANH	<3.3	✓ Ready · below detection, limit 3.3
PRICE, ELI	<5	⚠ Loads, flagged: detection limit 5 is above the child case level 3.5
HAYES, NORA	2.4	✓ Ready
REED, MILO	6.8	! Would be held: missing DOB

35 ready | 1 flagged | 2 would be held

(5) Saving stores the file's header signature with version 1. A later file with different columns pauses before loading.

Save profile, version 1 | Back to file settings

(1) Layout drift pauses
When a sender's columns change, the whole batch stops before loading and asks for a mapping update. A paused batch resumes once fixed; it is not a failed batch.
Intake profiles › Layout drift

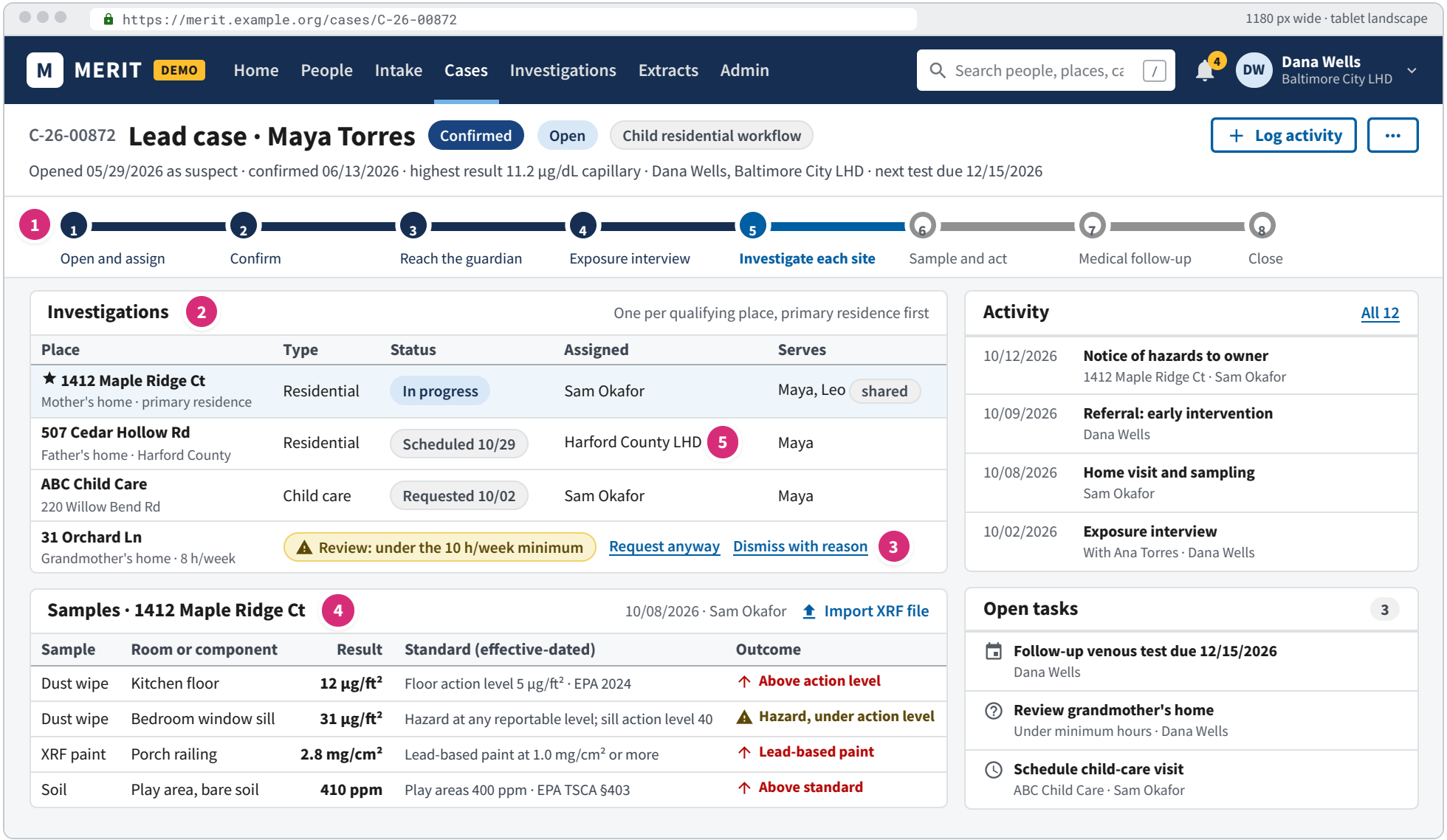
(2) Sample values beside fields
Each column shows real values from the uploaded sample, with suggested targets drawn from header names and content such as dates and ZIP codes.
Intake profiles › Building one

(3) Translations, codes, constants
Value translations, local test codes mapped to LOINC (which can imply the sample type) and constants for values the file never sends.
Intake profiles › What a profile holds

(4) Preview before saving
The first rows run through validation. A "<5" result loads but joins the detection-limit queue, because 5 is above the child case level.
Intake profiles · Work queues

(5) Versioned profiles
Each version keeps its header signature. A batch loaded with a bad mapping can be reprocessed with the corrected version.
Intake profiles › Versioning

Case and investigations



1 Workflow as steps
The child residential workflow's eight steps appear as a USWDS step indicator; each completed step links to the activities that closed it.
Child: residential workflow

2 One per place, reused
Each qualifying place gets one investigation, primary residence first. Leo's case joined the open one at his mother's home instead of starting another.
Investigation · Address rule 11

3 Below the minimum, a review
A place visited under the configured hours per week gets a review task. Staff can still request an investigation or dismiss it with a reason.
Address rule 11 · Example: shared custody

4 Outcomes are computed
Each sample is compared with an effective-dated standard, here EPA's 2024 dust rule, and "exceeds" is calculated rather than typed. XRF readings import through an intake profile.
Investigation › Environmental samples

5 Across tenants
The father's home is in Harford County. Work that spans tenants goes to their nearest common ancestor, which shares it with Harford County LHD.
Cross-tenant work

Threshold administration

https://merit.example.org/admin/thresholds/lead-blood

1180 px wide · tablet landscape

MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | ME Morgan Ellis, State lead program

Jurisdiction settings | Analytes | Thresholds | Environmental standards | Workflows | Matching | Extract layouts | Users and profiles | Audit log

Thresholds

Analyte specimen: **Lead · whole blood** | Root tenant setting · inherited by 25 child tenants | History | + Add threshold

New results are evaluated under the rows in effect. Open and closed cases keep their classification; a one-time report lists people a new level would reach, for outreach. (4)

(1) Population: **All** | Type: **All** | ✓ Show inactive reference rows | 9 rows

Type · population · sample	Level	Triggers	Effective from	Authority	Status
Reportable · all · any	**any value**	Report required	01/01/2020	State regulation	Active
Case · child · any	**≥ 3.5**	Opens a case	01/01/2023	CDC reference value	Active
Action · child · capillary (2)	**≥ 3.5**	Venous test in 3 months	01/01/2023	CDC guidance	Active
Action · child · capillary	**≥ 10**	Venous test in 1 month	01/01/2023	CDC guidance	Active
Action · child · capillary	**≥ 20**	Venous test in 2 weeks	01/01/2023	CDC guidance	Active
Action · child · capillary	**≥ 45**	Venous test in 48 hours	01/01/2023	CDC guidance	Active
Case · adult · any	**≥ 5**	Opens a case	01/01/2026	NIOSH elevated level	Active
Case · adult · venous	**≥ 3.5**	Opens a case	not set	CSTE 22-EH-01	Reference (3)
Case · pregnant · any	**≥ 5**	Opens a case	01/01/2024	CDC 2010 guidelines	Active

Levels in µg/dL, compared in canonical units. A result takes the highest tier it meets.

Turn on a reference row

Case · adult · venous · ≥ 3.5 µg/dL

Shipped inactive with its authority: CSTE 22-EH-01, the 2023 national case definition.

Effective from * 01/01/2027 | Comparator ≥

☐ Also end the NIOSH 5 µg/dL adult row on 12/31/2026

(5) **214** adults had a venous result from 3.5 to 4.9 µg/dL in the last 12 months

They are not reclassified. Run the outreach report to contact them.

Turn on from 01/01/2027 | Cancel

(1) Rules are dated data
Every level carries its population, sample type, comparator, effective dates and authority, so staff can see where a number came from and when it started.
Threshold · Design principle 4

(2) Tiers side by side
CDC's confirmatory-test schedule is four action rows, each with its own due interval. A capillary 12.4 meets the 10 tier: venous test within a month.
Threshold › action levels

(3) Reference rows ship inactive
Federal levels come with the code but stay off until an adopter turns them on. State levels are never shipped as defaults.
Threshold · Reference data shipped with the code

(4) No silent reclassification
When a level changes, open and closed cases keep their classification and new results use the new rows. A one-time report lists people whose recent results would now qualify.
Case classification › General rules

(5) Root-tenant setting
Thresholds belong to the root tenant and are inherited down the tree. Each change is versioned and audited, and administrator profiles do not include case editing.
Data ownership · Profiles and permissions

Federal extracts

https://merit.example.org/extracts — 1180 px wide · tablet landscape

M MERIT DEMO | Home | People | Intake | Cases | Investigations | Extracts | Admin | Search people, places, ca / | 4 | JR Jamie Rivera, State lead program

Federal extracts

Generated, validated and stored on a schedule · State lead program

Schedules and layouts

1

CBLS 2026 Q3 · children

Validated with 3 warnings

Quarterly · collected 07/01–09/30/2026 · ASCII fixed-width · CDC variable specification

Generated 10/05/2026 06:00 by schedule · stored with its validation report

31,284	28,906	0	3
records	children	errors	warnings

Review warnings | Download file | Mark submitted

ABLES 2026 · adults

Dry run: 4 rules failing

Annual · collected 01/01–12/31/2026 · final file after year end

2

Layout: NIOSH ABLES submission, April 2024 · OMB 0920-0931, exp. 07/31/2027

1,146	41	57	10
adults	status 1, new case	status 2, existing case	records failing

Fix records | Run dry run again

3

ABLES 2026 dry run · validation

Export report

NIOSH cross-field rule	Records	Example	Fix in
NAICS needs a valid code or 999 when WorkRel is 1 (work) or 3 (both)	4	P-0052-0117 · employer has no NAICS	Organization
COC needs a code or 990 (unknown) when WorkRel is 1 or 3	3	Marcus Bell · job not coded	Employment
Process must be NA when WorkRel is 1 or 9 (unknown)	2	P-0047-2290 · source text present	Case exposure
NAICS must be blank when WorkRel is 2 (not work) or 9	1	P-0031-7764 · employer code on a hobby case	Case exposure

5

Submissions

Audited, with row counts

CBLS 2026 Q2 — Submitted
07/14/2026 · Jamie Rivera · 30,977 records

CBLS 2026 Q1 — Submitted
04/13/2026 · Jamie Rivera · 29,412 records

ABLES 2025 — Submitted
03/18/2026 · Rosa Chen · 1,089 adults

4

Never in either file: names, street addresses, phone numbers, SSNs or medical record numbers. Records carry a stable surrogate ID, date of birth and the reporting county.

1 Built on a schedule
Both files are generated, validated and stored by the system, with a validation report before anyone submits: CBLS quarterly for children, ABLES once a year for adults.
Federal extracts

2 Layouts are versioned data
Each file names the layout it was built to. When NIOSH or CDC revise a layout, a new effective-dated version is loaded rather than code changed.
Design principle 4 · Jurisdiction configuration

3 NIOSH's own cross-field rules
A dry run before year end catches records that would fail, such as a work-related exposure without NAICS or occupation codes, and links to where each is fixed.
Federal extracts · Employment

4 Minimum necessary
Direct identifiers never leave in either file; each carries only the fields its federal layout requires.
Federal extracts · Privacy and network rules

5 Submission is a permission
Marking a file submitted needs extract.submit, and every export and submission is audited with its row count.
Profiles and permissions · General